Special Section: Arguing Digital History

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Arguing with Digital History: Patterns of Historical Interpretation

Abstract

Digital history has only rarely created interpretative or argumentative scholarship that contributes back to disciplinary understandings of the past. The reasons for this are varied: digital historians have often preferred to create scholarship for public audiences, or they have pursued forms of scholarship which do not lend themselves to explicit interpretation. But we contend that digital historians have not had sufficient models of patterns of historical argumentation compatible with digital historical research. In this introduction, we read the articles by Rachel Midura and Leonardo Barleta that appear in this special section, and a series of other articles, in order to show the patterns of historical argumentation that digital historians can pursue. A companion website features versions of the articles by Midura and Barleta, as well as eight additional previously published articles, annotated by their authors to highlight how they developed their historical arguments so they can serve as models for argumentative digital history.

Digital history has only rarely created interpretative or argumentative scholarship in the ways that currently define both the forms and the ends of disciplinary practices of research. In terms of form, digital historians have only rarely published the interpretative (rather than methodological) journal articles or monographs that are the main mediums for research in the discipline. Perhaps this is not surprising, given that the possibilities of digital media have spurred digital historians to creativity in coming up with new forms of scholarship.

But form leads to substance. Whatever their shortcomings, the journal article and the monograph excel at disciplining historians' research so that it yields argumentative scholarship that advances new interpretations of a particular field of study. That is the end to which most conventional historical research points. Digital history's creativity with form has not been matched by a corresponding impact on how historians understand the past.[1] At times some digital historians have wondered whether their methodologically defined field ought to be making more of an interpretative impact. More often, they have celebrated—or at least justified—their freedom from being constrained to pursue the same end as other historians. Certainly, digital history has spurred a great deal of methodological

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discussion and has brought historical knowledge to public audiences and to K–12 teachers, students, and families. Each of these is an end worth pursuing.

But it is our view that digital history should—and can—make more of an impact on historical knowledge within specific fields. Historical interpretations and arguments are not the only ends that digital history should pursue, but they are *an* end that digital history should pursue. For several years we have sought to encourage and enable other digital historians to pursue such argumentative and interpretative scholarship. The outcomes have included an Andrew W. Mellon Foundation–funded workshop that produced a white paper on historical argumentation in digital history as well as a journal that publishes discipline-specific arguments and interpretations that come out of digital history projects in progress. This special section, the product of another Mellon-funded workshop that brought together early-career historians and the editors of this journal, represents another piece of this ongoing project. The section is accompanied by a website hosted by the Roy Rosenzweig Center for History and New Media in which authors of noteworthy journal articles, including the two articles that appear here, annotate the process by which they developed their historical arguments in order to explain how they can be models for argumentative digital history.

At the same time, other historians have been slowly amassing a body of interpretative work based on digital history. In this introduction, we offer a brief overview of the reasons that digital historians have not in the main pursued interpretative scholarship. Then we turn to an examination of several cases in which digital historians have made such arguments, including in this special section, and seek to show the patterns of argumentation that those articles have developed.

Our argument is straightforward. Conventional historians have many models of scholarship in forms that pursue the end of historical argumentation. Digital historians have many models of scholarship in forms that encourage other ends, but they have few models of argument-driven digital history. Therefore it is not obvious to scholars in the field, especially early-career researchers, that they can pursue historical interpretation as a primary end or how they would go about creating such arguments. In this introduction, we seek to show how other digital historians have pursued digital scholarship that is methodologically innovative but also advances historical interpretations. By showing the patterns of argumentation latent in that scholarship, we seek to encourage other digital historians to emulate those patterns and create more interpretative scholarship.

Why Have Digital Historians Not Made Arguments?

Why digital history has not had the impact on field-specific arguments in history—an impact on the historiography—remains something of a puzzle. Certainly, there has been no shortage of claims by digital historians that they *would* have such an impact.

However, a large proportion of digital historians have instead concentrated on speaking directly to or with public audiences. In part that reflects the orientation of many early digital historians toward public history and the influence of the wealth of resulting models for such scholarship.[2] Yet even beyond public history, digital historians have made their work accessible to nonacademic

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audiences to a greater extent than have their non-digital colleagues. The products of digital research and analysis can be made available online to a far greater extent than is the case with analog history. Historical sources that have been collected and digitized, data created from sources, and exploratory visualizations and maps can all be shared with the public without making scholarly arguments and without the constraints of scholarly publication. (We ourselves have taken this approach with digital research, most notably in *Digital Harlem*, in the case of Stephen Robertson, and in *America's Public Bible*, in the case of Lincoln Mullen.) Many digital historians find audiences which are far larger than the audience of academic historians in a particular field, and they also feel that making historical resources available for open use is more rewarding and more valuable than the contribution of another journal article or monograph. In the past, such activity was seen as a kind of scholarship; with his typical prescience, Roy Rosenzweig recognized almost twenty years ago that the digital era would unsettle the relationship between historians and archival work and force us to ask, "Should the work of collecting, organizing, editing and preserving of primary sources receive the same kind of recognition and respect that it did in the earlier days of the profession?"[3]

The re-emergent digital forms of that practice among historians have slowly gained institutional and professional recognition, even if conventional historians too often accord it a standing that is short of the status of scholarship. Creating digital historical resources is also labor intensive, especially at scale, so digital historians who undertake it are left little time to build such an argument.

Moreover, often the exploratory analyses and visualizations that digital historians make public were never intended to be developed into argument-driven scholarship. Instead, the purpose was to find out what digital tools could do, to experiment. Tom Scheinfeldt succinctly articulated that approach in a blog post in 2010, arguing that only after "tool building, experimentation, and description" would digital tools be "sufficiently articulated and phenomena sufficiently described" for arguments to be made.[4] These experiments typically focused on working with data by using a digital tool to identify features and patterns in those sources, rather than answering questions or organizing the results into an overarching argument. The work of Ben Schmidt, for instance, affords many fascinating examples of visualizations intended to explore historical sources and illustrate the potential for data analysis as a method. His map of ship paths, for example, was created "to illustrate how rich metadata alone can be as a source for historical research."[5]

In cases where public digital historical resources have become the bases of scholarly arguments, they have had an indirect impact as the basis for argumentative history without being the interpretation itself. One need look no further than one of the earliest and most influential digital history projects: *The Valley of the Shadow*.[6] The sources the site contains form the basis of Edward Ayers's award-winning books *In the Presence of Mine Enemies* and *The Thin Light of Freedom*.[7] While a page in the front matter of the first volume and a mention in the preface of the second volume identifies the books as a part of *The Valley of Shadow* project and points readers to the site, the books themselves offer a narrative history that shows no indication of its digital roots and no explanation of how its argument is based on digital methods. To take another example, Jane Kamensky's *The Exchange Artist* is a contribution to the history of the early

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American republic, and most scholars have recognized the virtues of its narrative and prose style. But *The Exchange Artist* also features plates of a 3D reconstruction of Boston's Exchange Coffee House, which in turn informs the prose and the analysis. Again, the book does not present itself as a work of digital history, but it is one, at least in part.[8]

It is also the case that argument-driven scholarship based on digital research is not always visible as digital history when the emphasis is on interpretation rather than on method. "Digital history" and especially "Digital Humanities" define themselves in an institutional sense by drawing attention to the digital against the conventional. Scholarship in Digital Humanities is noticed when there is something about its approach or method that appeals across the typical boundaries of a field or discipline. A kind of map, or maybe an approach to text analysis, might draw the attention of a twentieth-century U.S. historian, an Ottomanist, and a literary scholar for its contribution to method. But if the work is pitched primarily as a contribution to Ottoman history, then the U.S. historian is unlikely to pay it much attention, and vice versa. Put differently, what draws the interest of attendees at Digital Humanities Summer Institute or the Digital Humanities conference is very different than what draws the attention of attendees at the Organization of American Historians or the Council on Latin American History.

One thing, however, is clear. The problem with digital history's impact on specific historical fields is no longer that editors in those fields are unwilling to publish such research. In recent years, leading journals such as the *American Historical Review*, the *Journal of American History*, the *Journal of Social History*, *Law and History Review* and others have shown that they are willing and even eager to publish scholarship in these fields, although not always in the form, or at least with the discussion of method, digital historians favor. If anything, digital history is at a premium for these journals, so long as it makes an argument recognizable as a contribution to disciplinary scholarship.

It is our contention that part of the reason this scholarship has not developed is because digital historians lack models for the kinds of work they can, and likely should, be doing. To that end, we have presented some models for how digital historians can connect their unique methodologies to historical interpretation.

Workshopping Arguments

A white paper produced by participants in a workshop we convened, "Digital History and Argument," provides a guide to the argumentative structures at work in digital collections, digital public history, digital methodological work, computational digital history, and visualizations.[9] The twenty-seven authors of the white paper concluded that most forms of digital history have advanced only implicit historical arguments.

Digital collections, on the one hand, are primarily about gathering and publishing primary sources. Scholars make an implicit argument by selecting, organizing, categorizing, and describing sources, but they do not explicitly articulate those arguments or address the implications of their collections for how the topic is understood by other scholars. The discussion of digital methodologies, on the other hand, explains and develops new ways to analyze historical sources,

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but again, those methods are not the interpretations themselves. Because digital public history is created both for public audiences and often in collaboration with those communities, it primarily engages in a conversation with the public rather than with other scholars, and it prefers implicit interpretation rather than explicit argumentation.

Computational digital history and digital history visualizations are more amenable to explicit argument, as they both create a broad view of a set of sources that highlights patterns that can be interpreted. Word frequency analysis, topic modeling, and word-embedded modeling identify quantitative patterns in textual data, while centrality and betweenness calculations identify quantitative patterns in networks. Maps and network visualizations identify spatial patterns in data. But projects involving both forms of digital history often present those quantitative analyses and visualizations with little interpretation. By contrast, the argumentative interpretation that appears in the pages of scholarly historical journals such as the *Journal of Social History* makes explicit its claims, evidence, logic, and engagement with other scholarship. Examining why there are so few examples of digital history that make such explicit arguments, the authors of the white paper concluded that scholars lack conceptual models of how to apply digital methods to historical questions.

To encourage more argument-driven digital history we collaborated with the Andrew W. Mellon Foundation to organize a series of workshops to support authors in the process of writing, peer review, and publication. The two articles that appear in this special issue/section are the result of that process. To extend the utility of these articles, we also created a site, *Models of Argument-Driven Digital History*, hosting a freely available version of the articles annotated to serve as models of how to conceive and construct interpretations and arguments using digital history methods and materials. When the changed circumstances of the pandemic curtailed plans to discuss the process and models in a session at the American Historical Association annual meeting, we moved instead to expand the site with eight additional articles, published scholarship that provides successful models of argument-driven digital history.[10]

Patterns of Argument-Driven Digital History

In this section we analyze a set of articles which have made field-specific arguments on the basis of digital historical work. In doing so, we seek to show two things. First, it is indeed possible for digital historians to make significant arguments in their historical fields. However, as we previously mentioned, such work often does not get marked as digital history because its contributions are interpretative rather than methodological, and because it is published in journals for specific fields rather than in Digital Humanities journals. So we have attempted to read broadly across fields other than our own to identify examples of digital historical arguments, though of course we have undoubtedly missed many worthy examples in other fields or languages other than English. Second, we have tried to read these articles to show how they can serve as models for other historians to make similar arguments. Advances in the historical discipline often occur when specific essays provide models for how other historians can offer new interpretations: one thinks of Robert Darnton's "The Great Cat Massacre" or Joan Scott's "Gender: A Useful Category of Analysis."[11] While

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perhaps none of the articles discussed here rises to the level of those celebrated works of scholarship, they are useful models for digital historians seeking to make arguments in different fields on the basis of similar kinds of sources and methods.

Published digital history that successfully makes explicit arguments shares several broad structures, with variations associated with different digital methods. The selection of evidence with which the process of historical argument begins is framed more broadly in digital history, including fragmentary sources not amenable to close reading and sets of sources expanded in scale and comprehensiveness. That foundation supports arguments centrally concerned with context, and that develop by moving across scales, often beginning with the broadest picture. (Here digital history differs from conventional historical arguments, which tend to move from the particular to its context.) Tim Hitchcock has described that as a "data-first" approach, in which the historian "purposefully presents a comprehensive set of data, on the basis of which the reader is guided to a specific conclusion" as opposed to the typical approach of the conventional historian, which centers "a claim or argument, evidenced through a narrow selection of precisely relevant data."[12] Digital methods provide an empirically grounded picture of the typical and representative in contrast to outliers, allowing for a systematic visual or quantitative examination of patterns through correlation, resemblance, or proximity. The sources selected through this process as the most important are the subject of close reading to address interpretive questions that the digital methods being used cannot answer, reengaging with the complexities of sources necessarily simplified to create data.

The forms of digital history most amenable to explicit disciplinary arguments are visualizations of data such as maps, networks and 3D models, and computational analysis of data. Far from mutually exclusive, those forms are often used in combination to produce network visualizations analyzed using algorithms and computational text analysis presented as maps and network graphs.

Digital historians developing arguments based on spatial visualizations are building on historians' long-standing use of mapping as a conceptual framework, albeit from a metaphorical perspective: the use of "the idiom of borders and boundaries, frontiers and crossroads, centers and margins," as Karen Halttunen put it.[13] Moreover, maps conveying information are a longstanding feature of historical publications, although generally they are limited to identifying locations. Digital history spatial visualizations by contrast show data in a spatial context, expanding the scale of information being conveyed, incorporating fragmentary sources, combining layers of different kinds of sources, and facilitating analysis at multiple scales. In employing qualitative sources and not relying on statistical analyses, these visualizations differ from mapping using Historical GIS. In "Seeing Emancipation: Scale and Freedom in the American South," Edward Ayers and Scott Nesbit mapped locations and movements of Union troops compiled by Frederick Dyer, and combined them with locations of what they defined as "emancipation events" gleaned from the Official Records of the War of the Rebellion, together with several newspapers and diaries. This map put together the pieces of the "vast, distended and chaotic" process of emancipation that unfolded within shifting boundaries and at an uneven pace to provide a picture that showed "the patterns, proportions and timing of emancipation," and "how actions overlap, penetrate, and conflict with each other."[14] Ayers and

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Nesbit progressively shift the scale of their analysis to examine specific patterns within the map, selecting first the entire Civil War South in summer 1864 to show how "military movements overlay legal and demographic geographies to create a complex terrain for emancipation," then narrowing the focus to Virginia to highlight "a loose coordination between enslaved men and women fleeing slavery and the paths of large armies," switching to a narrative that shows that "armies were unreliable vehicles for emancipation, bringing heartbreak as well as liberation."[15] In this argument digital mapping provides an empirically grounded, more comprehensive picture of emancipation and its relationship to the presence of Union troops, making visible patterns in the proximity of the two types of events that are then explored at different scales to show the impact of varying contexts and the deeply contingent nature of emancipation.

A further form of argument based on digital mapping uses quantification to identify patterns beyond those made visible on the map.[16] Harmony Bench and Kate Elswit, in their "Katherine Dunham's Global Method and the Embodied Politics of Dance's Everyday," quantify movement and examine it at multiple scales. By using performance contracts, receipt books, personal logs and activity diaries, programs, newspaper clippings, and personal and professional correspondence to reconstruct where Dunham was every day between January 1, 1950 and December 31, 1953, they generated an analysis "counter to dominant historical approaches that take a midfield view to build broad narratives through a small set of exemplary moments, anecdotally illustrating an argument."[17] Their mapping of travel between locations and timeline of the duration of stays reveal patterns in touring that "rebalance the geography of Dunham scholarship," revealing the importance of cities beyond those featured in existing scholarship and of nightclubs as performance venues, new contexts in which to interpret her choreographic and ethnographic work. Bench and Elswit then turn to close reading to highlight the embodied labor involved in the movement, which a visualization of points connected by lines tends to obscure. They use the paradigm of friction to elaborate the difficulties Dunham encountered moving, including available modes of transportation, the logistics of crossing borders, racial restrictions on accommodations, and the constant need to exchange currencies. Those pressures wore not only on Dunham's mobility, but on her body. Granular data that reveal Dunham performing or rehearsing 74 percent of the days in the period under study, and traveling for more than half of the remaining time, directs attention to the wear and tear on Dunham's body and the fatigue and physical complaints that she wrote about. As in Ayers and Nesbit's argument, mapping here provides an empirically grounded, more comprehensive picture, with the additional use of quantification of the number and duration of visits to locations to identify previously unrecognized patterns that are then analyzed. In also exploring those patterns at different scales, Bench and Elswit go closer than Ayers and Nesbit to examine the human scale, an individual's experience of the movement visualized on the map.

The 3D reconstruction also provides the basis for arguments about movement at the human scale. The historian using that method locates individuals within a digital model of a space and then analyzes archival evidence of their activity in that context. In "Reconsidering Poor Law Institutions by Virtually Reconstructing and Re-Viewing an Eighteenth-Century Workhouse," Susannah Ottaway and Austin Mason created a model of the House of Industry at

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Gressenhall, Norfolk, that allowed "human-scale exploration using a first-person controller in the mode of commercial video games." They then deployed this model as a "virtual 'stage' for [their] historical actors."[18] Their argument develops by first examining the features of the building and then tracing the routes of the poor and guardians as they made their way to workrooms, dining spaces, the schoolroom, committee room, and attendant spaces. Ottaway and Mason show that the building reinforced paupers' ease of movement in and out of the workhouse and authorized the role and power of the guardians articulated in documents. At the same time, the building facilitated a mixing of different categories of the poor that was at odds with the authorities' commitment to the segregation of groups. This combination reveals that the building embodied how "new ambitions around policing were actively restrained by the expectations and actions of the poor in the last decades of the eighteenth century."[19] While creating a context is the foundation of this argument, it is employed differently here than by Ayers and Nesbit and Bench and Elswit; additional evidence is put in the context of the model, rather than patterns being identified within it. The argument develops by exploring the relationship between evidence of different activity and movement and the building model to show the gap between ideal and practice.[20]

Leo Barleta's article in this special issue, "Spatial Genealogies Mobility, Settlement, and Empire-Building in the Brazilian Backlands, 1650–1800," takes a different approach to patterns of movement, visualizing journeys as a network of relationships that highlights that kinship not geography or infrastructure was key to mobility.[21] His sources are genealogical compilations, containing thousands of brief biographies, fragmentary sources mined for places of birth and marriage that could be used to trace mobility.[22] Counter to arguments that emphasize the exceptionality of the journeys of intrepid explorers, Barleta's visualizations highlight the repetition and routinization of movement between coast and interior, with the thickness of the lines linking points indicating the volume of traffic (which Bench and Elswit visualized as a timeline). The way that the lines fill the resulting image visually conveys his argument that the space was connected and made cohesive through family ties, that "these colonists' sustained mobility wove together an otherwise fragmented territory." To elaborate the texture and social aspects of the lives captured in these patterns, Barleta turns to snippets of an individual life story, constructed from other sources. The argument thus follows the structure of establishing a context of journeys that makes visible patterns that are then analyzed.

Networks, like maps, have been widely used metaphorically by historians as a conceptual framework. The language of networks is applied, as Ahnert et al. note, to "communities of practitioners, the dissemination of ideas, or the relationship between certain texts, images or artifacts."[23] Digital historians' network visualizations show subjects in the context of relationships, often with a secondary use of computational measures to identify for closer analysis those highly connected and those who by connecting the most individuals are important to the structure of the network. In "Kaskaskia Social Network: Kinship and Assimilation in the French-Illinois Borderlands, 1695–1735," Robert Morrissey uses a network created from marriage and baptismal records to understand the function over time of marriages between French men and Native American women in Illinois country in the early eighteenth century and "the communities

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and identities they sometimes created."[24] Morrissey develops his argument through a series of network visualizations for two time periods, using the patterns that are made visible to identify highly connected individuals for close analysis. In the first period patterns can be relatively easily seen in a graph of godparents and marriages, and a graph limited to godmothers, with directional links from godmother to parents. The most highly connected men and women in the network were those most oriented toward the agrarian life considered French. To more clearly see the social prominence and influence of leading families he switches to computational measures of centrality, betweenness scores (how many people are connected through a certain individual) combined with additional quantitative data about the land they had under cultivation and their household size. The results of that analysis show that the most connected and those that connected the most individuals had the greatest involvement in farming. The network's picture of kinship ties exerting pressure toward French agrarian cultures is very different from a historiography that argues that mixed race fur trade marriages were instead "rooted in an indigenous framework and an indigenous sphere."[25] In this argument the network graph provides a comprehensive context of relationships, making visible patterns of connections. Quantitative measures are then used to identify patterns within those connections in terms of who was connected to the most individuals and who connected the most individuals.[26]

Arguments focused on complex relationships rely more heavily on quantitative measures. The network of epistolary data that Ruth Ahnert and Sebastian Ahnert develop in their "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach" included, in addition to sender and recipient, other relationships mentioned in the correspondence: requested links, implied links, reported links, messenger links, spousal links, and other family links.[27] While the whole network graph confirms the importance of the martyrs as hubs in the community, plotting the taxonomy of links identifies another large group of people who are highly connected despite sending or receiving relatively few letters. To understand the role of such individuals in the network, Ahnert and Ahnert calculated not just those who connected the most individuals but also how many well-connected individuals an individual was adjacent to in the network (a measure called eigenvector centrality). High scores in those measurements identified men, and more surprisingly women, who provided the infrastructure of the network by carrying letters and delivering financial support. To bring this pattern further into focus, "to identify general rules for the overall structure and function of the network," Ahnert and Ahnert next developed quantitative criteria to establish categories of roles in the network, grouping them into three categories of "leader," three categories of network "sustainer" (including financial supporters and letter couriers), and one for a peripheral figure. A temporal examination of the network highlights that it was able to endure a systematic attack that removed fourteen of the twenty most central individuals thanks to the survival of these lesser-known figures who provided its infrastructural backbone. While the broad structure of this argument echoes that of Morrissey in establishing a context of relationships, it relies more on quantitative calculations to identify previously unrecognized patterns for analysis and to analyze the role in the network of the individuals prominent in those patterns.

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Rachel Midura's article in this special section, "Itinerating Europe: Early Modern Spatial Networks in Printed Itineraries, 1545–1700," uses networks to analyze itineraries, a fragmentary source that cannot be effectively analyzed with close reading. Since these documents contain geographical information, mapping might seem to be the appropriate form of visualization, but itineraries "presented cities as semi-spatial, abstract concepts and connections."[28] The abstract space of the network is a better fit than the geographical space of the map, preserving "the elements of hierarchy, directionality, and centrality conveyed by the original itinerary format."[29] Midura has created a network that places routes in context, allowing her to visually and quantitatively identify anomalies and demonstrate long term trends.[30] The relative importance of cities within the itinerary corpus is first analyzed visually by examining the number of connections to locations in the whole network. Following the structure of argument common to network graph analysis, Midura turns to centrality metrics for a more nuanced approach, calculating both betweenness and eigenvector scores. The results are visualized on a map to guide the analysis in the direction of real journeys, as each measure highlights a different feature of the routes: their inclusion of pilgrimage routes and the structuring effect of the mountain passes. However, adding a temporal dimension, moving to a dynamic network model, reveals that both inherited pilgrimage and mercantile routes "played a diminishing structural role in the overall conceptual network."[31] The life of a route was based on the period of its active publication in itineraries; analysis of route data and close reading identifies four clear periods, which are refined by creating a dynamic network model and extracting static networks to represent slices of time. Tracing the changes in the network, Midura finds the most dramatic shift near the end of the period, when "the cities included, the types of connections between them, and the overall structure of the conceived European systems of communication, travel, and exchange were altered."[32] Midura's argument comes out of making a connection between the historical phenomenon she studies—travel and the relationship between cities—and the way that phenomenon can be measured, visualized, and above all modeled by the networks she creates. As with other network-based arguments, Midura creates a broad context of relationships, in this case between cities, and then uses quantitative measures to identify and analyze patterns in that visualization. As she points out in the annotated version of the article, that analysis of the graph "relies on a spectrum of interpretations and visualizations, matching the approach to the question being asked in each section."[33]

Digital historians developing arguments based on computational text analysis lack an existing conceptual framework of the kind available for spatial visualizations and network graphs. The labels "distant reading" and "machine reading" suggest a relationship between computational text analysis and "close reading," an approach commonly used in humanities research. However, where digital maps and network graphs visualize data related to the concepts historians have used metaphorically, the results of computational text analysis have no such connection with close reading. The creation of data breaks text into words, with little or no context depending on how many consecutive words are grouped together. Since context is central to determining the meaning of words, the patterns found in the quantitative data, whether presented in charts, maps, or networks, in the first instance are not related to meaning. Text analysis thus

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relies more heavily than other digital methods on close reading to connect patterns with meaning.[34] As a result, this method is generally used to examine large collections of texts—more material than an individual could analyze with close reading.

One result of beginning with large datasets has been arguments focused on texts as objects of study, which begin with "a corpus looking for a question." Tim Hitchcock and William Turkel used word counts to reexamine the evidence the *Old Bailey Proceedings* provides of courtroom procedure; historians, based on a combination of close reading and statistical sampling, have assumed the length of trial reports reflects legal practices, such as very quick trials.[35] Hitchcock and Turkel analyzed the entire *Proceedings* as a "single massive text object," beginning by charting the number of words per year and trials per year. Read in relation to the existing literature, the chart generally fits conventional pictures of the eighteenth century, but is at odds with those of the nineteenth century, pointing to a series of specific moments that deserve further investigation. Shifting to an analysis of the *Proceedings* as a collection of trials, Hitchcock and Turkel analyze charts that show spasmodic changes in distribution of eighteenth-century trial lengths that indicate that they changed in response to factors outside court business. As such, they have limited value as evidence for the rise of the adversarial trial. By contrast, nineteenth-century trials show a gradual pattern of change and consistent pattern of reporting that suggests that they do reflect courtroom practice, and fit arguments about the rise of plea-bargaining in this period. Hitchcock and Turkel then examine specific factors associated with trials reported at different lengths: the seriousness of the offense, the verdict, and conviction rates. That quantitative analysis, possible because the digital text of the *Proceedings* is marked up with tags that identify information about those factors, confirms the two different regimes of trial reporting and the impact of the rise of plea bargaining. In using a comprehensive source to understand the nature of the source itself, not as a context for patterns visible within it, this form of argument focuses on identifying broad patterns across time rather than analyzing their details. It develops by applying different lenses that capture various facets of the data rather than shifting scale to close reading of material that forms patterns. As with Midura's argument, Hitchcock and Turkel were able to make a successful argument because they identified something about their dataset that could be measured (the word count of trials) which was a reasonable proxy for a historical change (the length and structure of legal proceedings), and then corroborated that change over time with other forms of historical knowledge.

A similar focus on identifying broad patterns shapes arguments that use topic modeling, the creation of groups of frequently collocated words, to identify some of the meaning of texts. Sharon Block and David Newman analyze tens of thousands of historical abstracts using a combination of word frequency and topic modeling to determine "what place does women's history have in the field at large and what kind of subjects are included within women's history; where in various regional histories does women's history most frequently appear; when do publications on women's history increase or decrease in numbers as well as when, chronologically, women's historians most focus their efforts."[36] The argument proceeds through three scales of analysis, calculating the amount of women's history in all the abstracts, and then using topic modeling to identify the

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subject areas in abstracts identified as women's history, concluding with a case study of one theme, the history of sexuality, with quantitative evidence of patterns presented in charts and tables. The broad pattern in which "different thematic emphases predominate in various regional specialties within women's history" is echoed in the case study (5 of the 40 topics, 9 percent of women's history abstracts), in which the amount and content related to sexuality varies more by region than chronology. Jo Guldi makes similar use of topic modeling to establish broad patterns in the parliamentary debates of Hansard—"new tensions and turning points that characterized the uptake of infrastructure over the longue durée."[37] In a model of 500 topics, a level that returned an overview of parliamentary business, Guldi looked for "words about technology, infrastructure, and the built environment, hoping to find patterns that replicate, enhance, or nuance the guns-maps-and-steel thesis of earlier scholars."[38] She chose a subset of topics and analyzed them as dynamic topic models, comparing the words and their relative ranking in the topic in different decades in tables shaded to highlight patterns. These topic models point to "significant contexts," "highlighting moments of shifting technology, institutional alignments, and political structures in the landscape."[39] Like arguments based on word frequency, these arguments based on topic modeling use a comprehensive dataset to understand the nature of the source itself, not as a context for patterns visible within it. They differ from arguments like that of Hitchcock and Turkel in developing their interpretation by shifting scale, to focus on particular topics rather than the text as a whole, and then on change over time, reflecting how topic models initially are not broken down temporally but need to be additionally analyzed as dynamic topic models.

A second strand of computational textual analysis approaches large datasets with specific questions and develops arguments that extend to close reading. Digital historians analyzing subjects that are denoted with proper nouns—people, places, organizations, products—have constructed arguments about those subjects based on word frequency. Cameron Blevins "measure[s] the frequency and distribution of specific geographic place-names across the pages of the *Houston Daily Post*" to examine "how newspapers construct space in an age of nationalizing forces."[40] While newspapers are a familiar source, Blevins's argument in "Space, Nation and the Triumph of Region" is based on evidence of a size and comprehensiveness made possible by digital methods; an examination of 1700 issues and aggregation of all the contents of those issues, from news to classified ads. He structures the argument in terms of the two distinct scales of the paper's imagined geography that appear when the frequencies with which places appear are visualized on maps: "a national scale oriented toward New York and the American Midwest, and a dominant regional scale of Texas and its immediate orbit." At the national scale, Blevins placed that "imagined" geography in the context of "actual" geography to highlight anomalies: cities that appear more than their population size would predict as well as an absence of the American South. Overlaying a map of the railroad network indicates that places that feature in the paper are oriented to that network. Blevins shifts the scale of his analysis to confirm that finding. First, he provides a close reading that reveals a recurring section of the newspaper devoted to railroads. Then a sampled content analysis, mapping what types of content appeared in what parts of each issue of the paper, demonstrates a larger relative presence of New York and

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Midwestern cities like Chicago in commercial content than any other category. At the regional scale, more of the newspaper focused on Texas and surrounding areas than further afield. Shifting scale to look at where places around Texas appeared, Blevins found them in fragmentary and non-narrative content such as advertisements, which came to prominence due to shifts in American journalism. While the large size of the set of sources used in this argument gives weight to its conclusion, the argument is based on a specific subset of the sources. In that way it differs from other patterns of argument in digital history that examine the relationships of patterns to the larger context, how typical or anomalous they are. Instead the argument looks for patterns within the place names he found and develops by shifting scale to close readings that identify the specific contexts in which those places appear, at the level of sections of the paper rather than within individual stories. Shift in scale fits Blevins' use of word frequency as a method for identifying the concept of space; studies focused on objects more often remain focused on overall patterns.[41]

Questions about borrowings and relationships between texts are generally framed in terms of specific publications, so they often involve not simply "a question looking for a corpus" but the creation of datasets for analysis. One of the authors, in collaboration with Kellen Funk, created a corpus of 222 statutes consisting of around 180,000 regulations to identify borrowings among those statutes and the extent of the influence of New York's Field Code. Dividing text into individual sections of code and working with five-word groups in each section, they generated similarity scores that identified from which section of a previous code, if any, it was most likely derived, identifying 106,000 borrowings. The argument of our "The Spine of American Law: Digital Text Analysis and U.S. Legal Practice" followed the structure of shifting scales toward a close reading, with different methods used to identify patterns at different scales of analysis: network analysis, visualizations, and clustering.[42] Examining each code as a whole, section-to-section borrowings were aggregated to show how many each code borrowed from each other code, with the results visualized as a network graph that showed the sequence of borrowing and creation of regional families of codes. Borrowing within each code was analyzed by visualizing each code as a grid with each box colored to the code from which it was borrowed. Finally, to analyze individual sections independent of their code, they were clustered based on similarity to each other and sorted chronologically. That approach of "taking them out of the context of the codes and putting them into the context of their particular variations" represented a form of "algorithmic close reading," directing attention to variations in sections.[43] These analyses identified the states whose archives should be the focus of research and close reading to explain the patterns of borrowing. Whereas arguments asking specific questions of a dataset using word frequency focus on only a subset of the data, this argument about text reuse is based on similarity scores and patterns within the data as a whole. Analysis of those patterns develops by progressively narrowing the scale, employing different visualizations at each scale that could be inspected both in the aggregate and in detail, and in which change over time is made visible by linking borrowings sequentially.

In conclusion, we think that the discussion above demonstrates that digital historians can make meaningful historical interpretations if they chose to do so. Certainly, despite earlier doubts that they would, digital historians have begun

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to make such arguments in their specific fields of study. By drawing together such works of scholarship for discussion in this essay, we hope to show the possibilities for argumentative scholarship to other digital historians.

But ultimately, digital historians will need to know not just whether such scholarship is possible but how they should go about it. The articles by Leonardo Barleta and Rachel Midura in this special section provide two models for such scholarship. And their annotated articles, along with those of eight other scholars that can found on the site *Models of Argument-Driven Digital History*, illustrate the patterns of argumentation implicit in their historical interpretations. In this introduction, we have attempted to identify successful digital history articles and also to explain the patterns of argumentation latent in those articles. One core pattern is argument through more comprehensive contextualization: historians put a person or place or text in the context of geography or network or corpus, or create a context in which to identify typical and anomalous examples and patterns. The transformation of sources into data required for digital tools means these contexts can be created out of fragmentary material. Other patterns of argument flow from creating a broad context of data. One pattern is argument through visualization and quantification: historians identify patterns in their sources by visualizing their data—patterns of spatial proximity in maps, connections in network graphs, and interaction with spaces in 3D models—or by using quantitative measures—calculations of who connected the most individuals in a graph, counts of word frequency, topic models of word collocation—or a combination of both. Another pattern is argument through scale: historians shift scales, exploring patterns in multiple contexts to identify their features and point to material that can be analyzed to interpret their meaning. A final pattern is argumentation through periodization: digital history that can show patterns in change over time that naturally connect to the arguments that historians make in periodizing their subjects.

Our point in listing out these patterns is not to create an exhaustive taxonomy of the ways that digital historians argue. Such a taxonomy would be as limited and rapidly out of date for digital history as it would be for conventional history. Rather, we suggest that thinking in terms of patterns of argumentation is a useful exercise for digital historians. Understanding the structure of arguments that other digital historians have made, including the many examples annotated on the companion website, should help digital historians think through the kinds of arguments that they themselves could make. We hope that many other digital historians will emulate such patterns and even develop new patterns of argumentation of their own.[44]

Endnotes

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tioned below. We thank them all for their willingness to take the time to share their work, and take on the challenge of developing historical arguments. Address correspondence to Stephen Robertson, George Mason University. Email: srober30@gmu.edu. Lincoln Mullen, George Mason University. Email: lincoln@lincolnmullen.com.

1. The exception is that, by and large, even conventional historical scholarship has been transformed by access to databases of primary sources. We are entirely in agreement with the point of view that searching databases is a form of digital scholarship. Nevertheless, we still find it worthwhile to draw a distinction between the practices of conventional historians, however much they have been affected by digital sources and research techniques, and the practices of digital historians who explicitly identify as such. See Lara Putnam, "The Transnational and the Text-Searchable: Digitized Sources and the Shadows They Cast," *American Historical Review* 121, no. 2 (2016): 377–402, https://doi.org/10.1093/ahr/121.2.377; Julia Laite, "The Emmet's Inch: Small History in a Digital Age," *Journal of Social History* 53, no. 4 (June 1, 2020): 963–89, https://doi.org/10.1093/jsh/shy118; and Tim Hitchcock, "Digital Searching and the Re-formulation of Historical Knowledge," in *The Virtual Representation of the Past*, ed. Mark Greengrass and Lorna Hughes (Farnham, 2008), 81–90.

2. Stephen Robertson, "The Differences Between Digital Humanities and Digital History," in *Debates in Digital Humanities 2016*, ed. Matthew Gold and Lauren Klein (Minneapolis, 2016), https://dhdebates.gc.cuny.edu/read/untitled/section/ed4a1145-7044-42e9-a898-5ff8691b6628#ch25.

3. Roy Rosenzweig, "Scarcity or Abundance? Preserving the Past in a Digital Age," *American Historical Review* 108, no. 3 (2003): 760.

4. Tom Scheinfeldt, "Where's the Beef? Does Digital Humanities Have to Answer Questions?" reprinted in *Debates in the Digital Humanities*, ed. Matthew K. Gold (Minneapolis, 2012), https://dhdebates.gc.cuny.edu/read/40de72d8-f153-43fa-836b-a41d241e949c/section/3c03ecdb-2dcf-4597-8fc4-e42f8dcc21e1#p1b2.

5. Ben Schmidt, "Ghost shipping paths" (n.d.): http://benschmidt.org/maps-visualizations-gallery/.

6. Edward Ayers, Anne Sarah Rubin, William G. Thomas III, and Andrew Torget, *The Valley of the Shadow: Two Communities in the American Civil War*, https://valley.lib.virginia.edu/.

7. Edward Ayers, *In the Presence of Mine Enemies: War in the Heart of America, 1859-1863* (New York, 2003); Edward Ayers, *The Thin Light of Freedom: The Civil War and Emancipation in the Heart of America* (New York, 2018).

8. Jane Kamensky, *The Exchange Artist: A Tale of High-Flying Speculation and America's First Banking Collapse* (New York, 2008).

9. Arguing with Digital History working group, "Digital History and Argument," white paper, Roy Rosenzweig Center for History and New Media (November 13, 2017), https://rrchnm.org/argument-white-paper/.

10. See *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

11. Robert Darnton, *The Great Cat Massacre and Other Episodes in French Cultural History* (New York, 1984); Joan Wallach Scott, "Gender: A Useful Category of Historical Analysis," *The American Historical Review* 91 no. 5 (1986): 1053–1075, https://doi.org/10.2307/1864376.

12. See "Data First" annotation, on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

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13. Karen Halttunen, "Groundwork: American Studies in Place: Presidential Address to the American Studies Association, November 4, 2005," *American Quarterly* 58, no. 1 (2006): 3.

14. Edward Ayers and Scott Nesbit, "Seeing Emancipation: Scale and Freedom in the American South," *Journal of the Civil War Era* 1, no. 1 (2011): 3, 10. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

15. Ayers and Nesbit, "Seeing Emancipation," 14, 17.

16. Scholars involved in the *Mapping the Republic of Letters* project used map-based visualizations of early modern correspondence networks as the basis for arguments that followed this pattern. Caroline Winterer drew on a map with locations sized for the number of letters sent there in "Where is America in the Republic of Letters?" to "reveal the hidden structures and conditions that nourished the growth of the republic of letters in the early modern period and the causes of its transformation in the nineteenth century," and provide a new context for that intellectual network and British America's place in it. See Caroline Winterer, "Where is America in the Republic of Letters?" *Modern Intellectual History* 9, no. 3 (2012): 597. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/. See also Dan Edelstein and Biliana Kassabova, "How England Fell Off the Map of Voltaire's Enlightenment article," *Modern Intellectual History* 17, no. 1 (2020): 29–53.

17. Harmony Bench and Kate Elswit, "Katherine Dunham's Global Method and the Embodied Politics of Dance's Everyday," *Theatre Survey* 61, no. 3 (2020): 308. For another example of an argument based on quantifying movement, in this case the voyages of shipping vessels, see Sean Fraga, "Digitally Mapping Commercial Currents: Maritime Mobility, Vessel Technology and U.S. Colonization of Puget Sound, 1851–1861," *Current Research in Digital History* 3 (2020), https://crdh.rrchnm.org/essays/v03-03-mapping-commercial-currents/.

18. Susannah Ottaway and Auston Mason, "Reconsidering Poor Law Institutions by Virtually Reconstructing and Re-Viewing an Eighteenth-Century Workhouse," *The Historical Journal* (2020): 10.

19. Ottaway and Mason, "Reconsidering Poor Law Institutions," 25.

20. That pattern of argument is not limited to projects that use 3D reconstruction. Scholars focused on cities have created more detailed base layers as a context for other data, so the relationship between specific places and events and individuals located there can be analyzed. For example, Nicholas Terpstra used DECIMA, a map that visualized census data from early modern Florence to provide a "socio-economic context" for three sets of data about "where prostitutes lived and worked, what prosecutions they faced most often, and what economic conditions marked their neighbourhoods." See Nicholas Terpstra, "Locating the Sex Trade in the Early Modern City," in *Mapping Space, Sense and Movement in Florence*, ed. Nicholas Terpstra and Colin Rose (Routledge, 2016), 107–124. See also Catherine Clarke, "Place, identity, and performance: spatial practices and social proxies in medieval Swansea," *Journal of Medieval History* 41, no. 3 (2015): 256–72.

21. Leo Barleta, "Spatial Genealogies Mobility, Settlement, and Empire-Building in the Brazilian Backlands, 1650-1800," *Journal of Social History*. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

22. More details of these sources and the process by which Barleta created data from them, see the annotation "About Sources" at *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

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23. Ruth Ahnert, Sebastian Ahnert, Catherine Nicole Coleman, and Scott Weingart, *The Network Turn: Changing Perspectives in the Humanities* (Cambridge, 2020), 7.

24. Robert Morrissey, "Kaskaskia Social Network: Kinship and Assimilation in the French-Illinois Borderlands, 1695–1735," *William and Mary Quarterly* 70, no. 1 (2013): 105.

25. Morrissey, "Kaskaskia Social Network," 142.

26. Network graphs created from more fragmentary sources produce variations on the structure of argument employed by Morrissey in addressing the nature of sources as well as the relationships evident in them. See Maeve Kane, "For Wagrassero's Wife's Son: Colonialism and the Structure of Indigenous Women's Social Connections, 1690–1730," *Journal of Early American History* 7, no. 2 (2017): 89–114. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/. For another example that uses a variety of different sources to identify the presence of women in Ottoman-Algerian social networks, see Ashley Saunders, "Silent No More: Women as Significant Political Intermediaries in Ottoman Algeria," *Current Research in Digital History* 3 (2020), https://crdh.rrchnm.org/essays/v03-05-silent-no-more/ .

27. Ruth Ahnert and Sebastian Ahnert, "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach," *ELH* 82, no. 1 (2015): 1–33. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

28. Rachel Midura, "Itinerating Europe: Early Modern Spatial Networks in Printed Itineraries, 1545–1700," *Journal of Social History*—page numbers to come? An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

29. Midura, "Itinerating Europe,"- page numbers to come?

30. Midura, "Itinerating Europe,"- page numbers to come?

31. Midura, "Itinerating Europe,"- page numbers to come?

32. Midura, "Itinerating Europe,"- page numbers to come?

33. Midura, "Itinerating Europe,"- page numbers to come?

34. For a useful elaboration of the relationship between distant reading (here termed machine reading) and close reading, see Katherine Hayles, *How We Think: Digital Media and Contemporary Technogenesis* (Chicago, 2012), 57–59, 68–75.

35. Tim Hitchcock and William Turkel, "The *Old Bailey Proceedings, 1674–1913*: Text Mining for Evidence of Court Behavior," *Law and History Review* 34, no. 4 (2016): 929–55. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

36. Sharon Block and David Newman, "What, Where, When, and Sometimes Why: Data Mining Two Decades of Women's History Abstracts," *Journal of Women's History* 23, no. 1 (2011): 82. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

37. Jo Guldi, "Parliament's Debates about Infrastructure: An Exercise in Using Dynamic Topic Models to Synthesize Historical Change," *Technology and Culture* 60, no. 1 (2019): 3. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

38. Guldi, "Parliament's Debates," 9.

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39. Guldi, "Parliament's Debates," 27.

40. Cameron Blevins, "Space, Nation, and the Triumph of Region: A View of the World from Houston," *Journal of American History* 101 1 (2014): 122–47; and Cameron Blevins, "Mining and Mapping the Production of Space: A View of the World from Houston," (2014), http://web.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id=93.

41. For example, Anne Helmreich, Tim Hitchcock, and William Turkel's analysis of objects listed as stolen in trials for theft uses not close reading but two-term phrases (bigrams) to "mitigate linguistic ambiguity by focusing attention on pairs typically constituted of an adjective and a noun, such as 'linen handkerchief." See Anne Helmreich, Tim Hitchcock, and William Turkel, "Rethinking Inventories in the Digital Age: The Case of the Old Bailey," *Journal of Art Historiography* 11 (2014), https://arthistoriography.files.wordpress.com/2014/11/helmreich_hitchcock_turkel.pdf.

42. Kellen Funk and Lincoln Mullen, "The Spine of American Law: Digital Text Analysis and U.S. Legal Practice," *American Historical Review* 123, no. 1 (2018): 132–64. For another example of an argument about text reuse developed at multiple scales, in this case in Indian treaties, see Joshua Catalano, "Digitally Analyzing the Uneven Ground: Language Borrowing Among Indian Treaties," *Current Research in Digital History* 1 (2018), https://crdh.rrchnm.org/essays/v01-02-digitally-analyzing-the-uneven-ground/. Melodee Beals develops a more iterative approach to shifting scales to explore the textual reappearance that characterized nineteenth-century newspapers. Her argument moves from a newspaper corpus to a close reading of five issues then scales the results of the close reading to reexamine the newspaper as a whole. See Melodee Beals, "Close Readings of Big Data: Triangulating Patterns of Textual Reappearance and Attribution in the *Caledonian Mercury*, 1820–40," *Victorian Periodicals Review* 51, no. 4 (2018): 616–39. An annotated version of this article appears on *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

43. Funk and Mullen, "The Spine of American Law," 154.

44. See *Models of Argument-Driven Digital History*, https://model-articles.rrchnm.org/.

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